

```
(kali㉿kali)-[~]
$ mkdir -p ~/CIP-B103-Lab6/{evidence,working,exported,reports,screenshots,scripts}

(kali㉿kali)-[~]
$ cd ~/CIP-B103-Lab6

(kali㉿kali)-[~/CIP-B103-Lab6]
$ pwd
/home/kali/CIP-B103-Lab6

(kali㉿kali)-[~/CIP-B103-Lab6]
$ find . -maxdepth 1 -type d -print
.
./working
./screenshots
./evidence
./scripts
./reports
./exported

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo apt install -y apache2 curl iptables tshark wireshark
apache2 is already the newest version (2.4.68-1).
curl is already the newest version (8.20.0-5).
```

The following packages were automatically installed and are no longer required:
libraw23t64 libvidstab1.1 openjdk-21-jre openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.

Upgrading:
iptables libip4tc2 libip6tc2 libxtables12

Summary:
Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 1738
Download size: 429 kB
Freed space: 8,192 B

```
Get:1 http://kali.download/kali kali-rolling/main amd64 iptables amd64 1.8.13-1 [361 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 libip4tc2 amd64 1.8.13-1 [18.6 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 libip6tc2 amd64 1.8.13-1 [18.9 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 libxtables12 amd64 1.8.13-1 [30.5 kB]
Fetched 429 kB in 1s (482 kB/s)
(Reading database... 432446 files and directories currently installed.)
Preparing to unpack .../iptables_1.8.13-1_amd64.deb...
Unpacking iptables (1.8.13-1) over (1.8.11-4)...
Preparing to unpack .../libip4tc2_1.8.13-1_amd64.deb...
Unpacking libip4tc2:amd64 (1.8.13-1) over (1.8.11-4)...
Preparing to unpack .../libip6tc2_1.8.13-1_amd64.deb...
Unpacking libip6tc2:amd64 (1.8.13-1) over (1.8.11-4)...
Preparing to unpack .../libxtables12_1.8.13-1_amd64.deb...
```

```
Setting up libip6tc2:amd64 (1.8.13-1)...
Setting up libxtables12:amd64 (1.8.13-1)...
Setting up iptables (1.8.13-1)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.1.5)...
Processing triggers for libc-bin (2.42-13)...
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-08-01 09:50:51 EDT; 1h 19min ago
 Invocation: 8f61f05317dd43fc9c8c68dcdcc8fa90
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 928 (apache2)
  Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:  0 B/sec"
    Tasks: 6 (limit: 2075)
  Memory: 27.3M (peak: 36.9M)
     CPU: 1.025s
   CGroup: /system.slice/apache2.service
```

```
CGroup: /system.slice/apache2.service
├─928 /usr/sbin/apache2 -k start -DFOREGROUND
├─980 /usr/sbin/apache2 -k start -DFOREGROUND
├─981 /usr/sbin/apache2 -k start -DFOREGROUND
├─982 /usr/sbin/apache2 -k start -DFOREGROUND
├─983 /usr/sbin/apache2 -k start -DFOREGROUND
└─984 /usr/sbin/apache2 -k start -DFOREGROUND

Aug 01 09:50:50 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 01 09:50:51 kali apachectl[928]: AH00558: apache2: Could not reliably determine the server's fully qualified
Aug 01 09:50:51 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ printf '<!DOCTYPE html>\n<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: FUSEINI IMORU KANTUOGAA</p></body></html>\n' | sudo tee /var/www/html/firewall_lab.html
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: FUSEINI IMORU KANTUOGAA</p></body></html>

(kali㉿kali)-[~/CIP-B103-Lab6]
$ cat /var/www/html/firewall_lab.html
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: FUSEINI IMORU KANTUOGAA</p></body></html>

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables-save | tee reports/iptables_before.rules
# Generated by iptables-save v1.8.13 (nf_tables) on Sat Aug  1 11:12:31 2026
*filter
:INPUT ACCEPT [0:0]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [0:0]
:DOCKER - [0:0]
:DOCKER-BRIDGE - [0:0]
:DOCKER-CT - [0:0]
:DOCKER-FORWARD - [0:0]
:DOCKER-ISOLATION-STAGE-1 - [0:0]
:DOCKER-ISOLATION-STAGE-2 - [0:0]
```

```

:DOCKER-ISOLATION-STAGE-2 - [0:0]
:DOCKER-USER - [0:0]
-A FORWARD -j DOCKER-USER
-A FORWARD -j DOCKER-FORWARD
-A DOCKER ! -i docker0 -o docker0 -j DROP
-A DOCKER-BRIDGE -o docker0 -j DOCKER
-A DOCKER-CT -o docker0 -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A DOCKER-FORWARD -j DOCKER-CT
-A DOCKER-FORWARD -j DOCKER-ISOLATION-STAGE-1
-A DOCKER-FORWARD -j DOCKER-BRIDGE
-A DOCKER-FORWARD -i docker0 -j ACCEPT
-A DOCKER-ISOLATION-STAGE-1 -i docker0 ! -o docker0 -j DOCKER-ISOLATION-STAGE-2
-A DOCKER-ISOLATION-STAGE-2 -o docker0 -j DROP
COMMIT
# Completed on Sat Aug  1 11:12:31 2026
# Generated by iptables-save v1.8.13 (nf_tables) on Sat Aug  1 11:12:31 2026
*nat
:PREROUTING ACCEPT [0:0]
:INPUT ACCEPT [0:0]
:OUTPUT ACCEPT [165:14270]
:POSTROUTING ACCEPT [165:14270]
:DOCKER - [0:0]
-A PREROUTING -m addrtype --dst-type LOCAL -j DOCKER
-A OUTPUT ! -d 127.0.0.0/8 -m addrtype --dst-type LOCAL -j DOCKER
-A POSTROUTING -s 172.17.0.0/16 ! -o docker0 -j MASQUERADE

```

```

-A POSTROUTING -s 172.17.0.0/16 ! -o docker0 -j MASQUERADE
-A DOCKER -i docker0 -j RETURN
COMMIT
# Completed on Sat Aug  1 11:12:31 2026

```

```

(kali@kali)~[~/CIP-B103-Lab6]

```

```

$ sudo iptables -L -n -v --line-numbers | tee reports/iptables_before.txt

```

```

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)

```

num	pkts	bytes	target	prot	opt	in	out	source	destination
-----	------	-------	--------	------	-----	----	-----	--------	-------------

```

Chain FORWARD (policy DROP 0 packets, 0 bytes)

```

num	pkts	bytes	target	prot	opt	in	out	source	destination
1	0	0	DOCKER-USER	all	--	*	*	0.0.0.0/0	0.0.0.0/0
2	0	0	DOCKER-FORWARD	all	--	*	*	0.0.0.0/0	0.0.0.0/0

```

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)

```

num	pkts	bytes	target	prot	opt	in	out	source	destination
-----	------	-------	--------	------	-----	----	-----	--------	-------------

```

Chain DOCKER (1 references)

```

num	pkts	bytes	target	prot	opt	in	out	source	destination
1	0	0	DROP	all	--	!docker0	docker0	0.0.0.0/0	0.0.0.0/0

```

Chain DOCKER-BRIDGE (1 references)

```

num	pkts	bytes	target	prot	opt	in	out	source	destination
1	0	0	DOCKER	all	--	*	docker0	0.0.0.0/0	0.0.0.0/0

```
Chain DOCKER-CT (1 references)
num  pkts bytes target    prot opt in     out     source    destination
1      0      0 ACCEPT    all  --  *      docker0  0.0.0.0/0      0.0.0.0/0
STABLISHED

Chain DOCKER-FORWARD (1 references)
num  pkts bytes target    prot opt in     out     source    destination
1      0      0 DOCKER-CT all  --  *      *        0.0.0.0/0      0.0.0.0/0
2      0      0 DOCKER-ISOLATION-STAGE-1 all  --  *      *        0.0.0.0/0      0.0.0.0/0
3      0      0 DOCKER-BRIDGE all  --  *      *        0.0.0.0/0      0.0.0.0/0
4      0      0 ACCEPT    all  --  docker0 *        0.0.0.0/0      0.0.0.0/0

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
num  pkts bytes target    prot opt in     out     source    destination
1      0      0 DOCKER-ISOLATION-STAGE-2 all  --  docker0 !docker0 0.0.0.0/0      0.0.0.0/0

Chain DOCKER-ISOLATION-STAGE-2 (1 references)
num  pkts bytes target    prot opt in     out     source    destination
1      0      0 DROP      all  --  *      docker0 0.0.0.0/0      0.0.0.0/0

Chain DOCKER-USER (1 references)
num  pkts bytes target    prot opt in     out     source    destination
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ sha256sum reports/iptables_before.rules | tee reports/iptables_before_sha256.txt
7ea21a99a87be47e44d1f2c5c3080811a780b3ad5051252ed07e07ed706fa7ad  reports/iptables_before.rules
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ ls -l reports
total 12
-rw-rw-r-- 1 kali kali 1312 Aug  1 11:12 iptables_before.rules
-rw-rw-r-- 1 kali kali  96 Aug  1 11:13 iptables_before_sha256.txt
-rw-rw-r-- 1 kali kali 2459 Aug  1 11:12 iptables_before.txt
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ tree
.
├── evidence
├── exported
├── reports
│   ├── iptables_before.rules
│   ├── iptables_before_sha256.txt
│   └── iptables_before.txt
├── screenshots
├── scripts
└── working
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ sha256sum reports/iptables_before.rules | tee reports/iptables_before_sha256.txt
7ea21a99a87be47e44d1f2c5c3080811a780b3ad5051252ed07e07ed706fa7ad  reports/iptables_before.rules
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ ls -l reports
total 12
-rw-rw-r-- 1 kali kali 1312 Aug  1 11:12 iptables_before.rules
-rw-rw-r-- 1 kali kali  96 Aug  1 11:13 iptables_before_sha256.txt
-rw-rw-r-- 1 kali kali 2459 Aug  1 11:12 iptables_before.txt
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ tree
.
├── evidence
├── exported
├── reports
│   ├── iptables_before.rules
│   ├── iptables_before_sha256.txt
│   └── iptables_before.txt
├── screenshots
├── scripts
└── working
```

7 directories, 3 files


```

(kali㉿kali)-[~/CIP-B103-Lab6]
$ ip -br address | tee reports/server_interfaces.txt
lo                UNKNOWN      127.0.0.1/8 ::1/128
eth0              UP          192.168.44.128/24 fe80::fcd8:733e:9de1:56cb/64
docker0          DOWN        172.17.0.1/16

(kali㉿kali)-[~/CIP-B103-Lab6]
$ ip route | tee reports/server_routes.txt
default via 192.168.44.2 dev eth0 proto dhcp src 192.168.44.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.44.0/24 dev eth0 proto kernel scope link src 192.168.44.128 metric 100

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ss -lntp | grep ':80' | tee reports/apache_listener.txt
LISTEN 0      511          *:80          *:~          users:((("apache2",pid=984,fd=4),("apache2",pid=983,fd=4),("apache2",pid=982,fd=4),("apache2",pid=981,fd=4),("apache2",pid=980,fd=4),("apache2",pid=928,fd=4))

(kali㉿kali)-[~/CIP-B103-Lab6]
$ ip -br address
lo                UNKNOWN      127.0.0.1/8 ::1/128
eth0              UP          192.168.44.128/24 fe80::fcd8:733e:9de1:56cb/64
docker0          DOWN        172.17.0.1/16

```

```

(kali㉿kali)-[~/CIP-B103-Lab6]
$ curl -v --connect-timeout 5 http://192.168.44.128/firewall_lab.html
* Trying 192.168.44.128:80 ...
* Established connection to 192.168.44.128 (192.168.44.128 port 80) from 192.168.44.128 port 43588
* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 192.168.44.128
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Sat, 01 Aug 2026 15:22:48 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Sat, 01 Aug 2026 15:11:43 GMT
< ETag: "7e-657fdb90e4afd"
< Accept-Ranges: bytes
< Content-Length: 126
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: FUSEINI IMORU KANTUOGAA</p></body></html>
* Connection #0 to host 192.168.44.128:80 left intact

```

```

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo tshark -i eth0 \
-f 'host 192.168.44.128 and tcp port 80' \
-a duration:30 \
-w /tmp/http_allowed.pcapng &
[4] 48010

(kali㉿kali)-[~/CIP-B103-Lab6]
$ Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

[4] done sudo tshark -i eth0 -f 'host 192.168.44.128 and tcp port 80' -a duration:30 -
(kali㉿kali)-[~/CIP-B103-Lab6]
$ curl -v http://192.168.44.128/firewall_lab.html
* Trying 192.168.44.128:80 ...
* Established connection to 192.168.44.128 (192.168.44.128 port 80) from 192.168.44.128 port 40690
* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 192.168.44.128
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off

```

```
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Sat, 01 Aug 2026 15:29:11 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Sat, 01 Aug 2026 15:11:43 GMT
< ETag: "7e-657fdb90e4afd"
< Accept-Ranges: bytes
< Content-Length: 126
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: FUSEINI IMORU KANTUOGAA</p></body></html>
* Connection #0 to host 192.168.44.128:80 left intact
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ ls -lh /tmp/http_allowed.pcapng
-rw-r--r-- 1 root root 368 Aug  1 11:29 /tmp/http_allowed.pcapng

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo cp /tmp/http_allowed.pcapng evidence/http_allowed.pcapng

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo chown kali:kali evidence/http_allowed.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo chown kali:kali evidence/http_allowed.pcapng

(kali㉿kali)-[~/CIP-B103-Lab6]
$ ls -lh evidence/http_allowed.pcapng
-rw-r--r-- 1 kali kali 368 Aug  1 11:29 evidence/http_allowed.pcapng

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sha256sum evidence/http_allowed.pcapng | tee reports/http_allowed_sha256.txt
36796ced54c306f845072897c3ccc85c3b5f6723b17fa59128fc8ecdffaa37bae  evidence/http_allowed.pcapng

(kali㉿kali)-[~/CIP-B103-Lab6]
$ tshark -r evidence/http_allowed.pcapng -Y http

(kali㉿kali)-[~/CIP-B103-Lab6]
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ ip -br address
lo                UNKNOWN          127.0.0.1/8 ::1/128
eth0              UP                192.168.44.128/24 fe80::fcd8:733e:9de1:56cb/64
docker0          DOWN             172.17.0.1/16

(kali㉿kali)-[~/CIP-B103-Lab6]
$ BLOCKED_CLIENT_IP=192.168.44.128

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables -I INPUT 1 -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables -L INPUT -n -v --line-number | tee reports/iptables_after_add.txt
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source               destination
1      0      0 DROP      tcp  --  *      *       192.168.44.128       0.0.0.0/0           tcp dpt:80

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables -C INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP

(kali㉿kali)-[~/CIP-B103-Lab6]
$ printf 'Rule verified present.\n' | tee reports/rule_verification.txt
Rule verified present.
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
```

```
$ tree
```

```
├── evidence
│   └── http_allowed.pcapng
├── exported
├── reports
│   ├── apache_listener.txt
│   ├── http_allowed_sha256.txt
│   ├── iptables_after_add.txt
│   ├── iptables_before.rules
│   ├── iptables_before_sha256.txt
│   ├── iptables_before.txt
│   ├── rule_verification.txt
│   ├── server_interfaces.txt
│   └── server_routes.txt
├── screenshots
├── scripts
└── working
```

7 directories, 10 files

```
(kali㉿kali)-[~/CIP-B103-Lab6]
```

```
$ █
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
```

```
$ sudo tshark -i eth0 \
-f 'host 192.168.44.128 and tcp port 80' \
-a duration:35 \
-w /tmp/http_blocked.pcapng &
[4] 53259
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
```

```
$ Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
```

```
[4] done sudo tshark -i eth0 -f 'host 192.168.44.128 and tcp port 80' -a duration:35 -
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
```

```
$ sleep 3
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
```

```
$ curl -v --connect-timeout 10 http://192.168.44.128/firewall_lab.html
* Trying 192.168.44.128:80...
* Connection timed out after 10006 milliseconds
* closing connection #0
curl: (28) Connection timed out after 10006 milliseconds
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo cp /tmp/http_blocked.pcapng evidence/http_blocked.pcapng

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo chown kali:kali evidence/http_blocked.pcapng

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_test.txt
Chain INPUT (policy ACCEPT 11 packets, 784 bytes)
num  pkts bytes target    prot opt in     out     source                 destination
1      7   420 DROP      tcp  --  *      *       192.168.44.128         0.0.0.0/0           tcp dpt:80
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ tree

.
├── evidence
│   ├── http_allowed.pcapng
│   └── http_blocked.pcapng
├── exported
├── reports
│   ├── apache_listener.txt
│   ├── http_allowed_sha256.txt
│   ├── iptables_after_add.txt
│   ├── iptables_after_test.txt
│   ├── iptables_before.rules
│   ├── iptables_before_sha256.txt
│   ├── iptables_before.txt
│   ├── rule_verification.txt
│   ├── server_interfaces.txt
│   └── server_routes.txt
├── screenshots
├── scripts
└── working
```

7 directories, 12 files

```
(kali㉿kali)-[~/CIP-B103-Lab6]
```



```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ip netns add client

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ip netns list
client

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ip link add veth-server type veth peer name veth-client

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ip link set veth-client netns client

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ip addr add 192.168.44.1/24 dev veth-server

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ip link set veth-server up

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ip netns exec client bash
(root㉿kali)-[/home/kali/CIP-B103-Lab6]
# ip addr add 192.168.44.129/24 dev veth-client

(root㉿kali)-[/home/kali/CIP-B103-Lab6]
# ip link set veth-client up

(root㉿kali)-[/home/kali/CIP-B103-Lab6]
# ip route add default via 192.168.44.1

(root㉿kali)-[/home/kali/CIP-B103-Lab6]
```

```

(kali㉿kali)-[/home/kali/CIP-B103-Lab6]
# ping -c 2 192.168.44.128
PING 192.168.44.128 (192.168.44.128) 56(84) bytes of data.
64 bytes from 192.168.44.128: icmp_seq=1 ttl=64 time=0.197 ms
64 bytes from 192.168.44.128: icmp_seq=2 ttl=64 time=0.123 ms

— 192.168.44.128 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1113ms
rtt min/avg/max/mdev = 0.123/0.160/0.197/0.037 ms

(kali㉿kali)-[/home/kali/CIP-B103-Lab6]
# exit
exit

(kali㉿kali)-[~/CIP-B103-Lab6]
$ ip addr show veth-server
5: veth-server@if4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
link/ether 32:99:9c:df:8f:79 brd ff:ff:ff:ff:ff:ff link-netns cli
inet 192.168.44.1/24 scope global veth-server
    valid_lft forever preferred_lft forever
inet6 fe80::3099:9cff:fedf:8f79/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo tshark -i veth-server \
-f 'host 192.168.44.129 and tcp port 80' \
-a duration:30 \
-w /tmp/http_allowed.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-server'
10

```

```

10

(kali㉿kali)-[~/CIP-B103-Lab6]
$ capinfos /tmp/http_allowed.pcapng | grep "Number of packets"
capinfos: You don't have permission to read the file "/tmp/http_allow
ed.pcapng".

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo chown kali:kali /tmp/http_allowed.pcapng

(kali㉿kali)-[~/CIP-B103-Lab6]
$ capinfos /tmp/http_allowed.pcapng | grep "Number of packets"
Number of packets:      10
      Number of packets = 10

(kali㉿kali)-[~/CIP-B103-Lab6]
$

```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ip netns exec client curl -v http://192.168.44.128/firewall_
[sudo] password for kali:
* Trying 192.168.44.128:80 ...
* Established connection to 192.168.44.128 (192.168.44.128 port 80) f
* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 192.168.44.128
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Sat, 01 Aug 2026 15:56:14 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Sat, 01 Aug 2026 15:11:43 GMT
< ETag: "7e-657fdb90e4afd"
< Accept-Ranges: bytes
< Content-Length: 126
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: F
html>
* Connection #0 to host 192.168.44.128:80 left intact

(kali㉿kali)-[~/CIP-B103-Lab6]
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ip netns exec client curl -v \
--connect-timeout 10 \
http://192.168.44.128/firewall_lab.html
[sudo] password for kali:
* Trying 192.168.44.128:80 ...
* Connection timed out after 10007 milliseconds
* closing connection #0
curl: (28) Connection timed out after 10007 milliseconds

(kali㉿kali)-[~/CIP-B103-Lab6]
$
```

```

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables -F INPUT

(kali㉿kali)-[~/CIP-B103-Lab6]
$ BLOCKED_CLIENT_IP=192.168.44.129

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables -I INPUT 1 \
-s "$BLOCKED_CLIENT_IP" \
-p tcp --dport 80 \
-j DROP

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables -L INPUT -n -v --line-number | tee reports/iptables_after_add.txt
Chain INPUT (policy ACCEPT 68 packets, 5487 bytes)
num  pkts bytes target    prot opt in     out     source               destination
1      0      0 DROP      tcp  --  *      *      192.168.44.129       0.0.0.0/0           tcp dpt:80

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables -C INPUT \
-s "$BLOCKED_CLIENT_IP" \
-p tcp --dport 80 \
-j DROP

printf 'Rule verified present.\n' | tee reports/rule_verification.txt
Rule verified present.

```

```

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo tshark -i veth-server \
-f 'host 192.168.44.129 and tcp port 80' \
-a duration:35 \
-w /tmp/http_blocked.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-server'
7

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo chown kali:kali /tmp/http_blocked.pcapng
cp /tmp/http_blocked.pcapng evidence/http_blocked.pcapng

(kali㉿kali)-[~/CIP-B103-Lab6]
$ capinfos evidence/http_blocked.pcapng | grep "Number of packets"
Number of packets: 7
Number of packets = 7

(kali㉿kali)-[~/CIP-B103-Lab6]
$

```

```

(kali㉿kali)-[~/CIP-B103-Lab6]
$ BLOCKED_CLIENT_IP=192.168.44.129

sudo iptables -D INPUT \
-s "$BLOCKED_CLIENT_IP" \
-p tcp --dport 80 \
-j DROP

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables -C INPUT \
-s "$BLOCKED_CLIENT_IP" \
-p tcp --dport 80 \
-j DROP || echo 'Rule successfully removed'
iptables: Bad rule (does a matching rule exist in that chain?).
Rule successfully removed

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo iptables -L INPUT -n -v --line-number | tee reports/iptables_restored.txt
Chain INPUT (policy ACCEPT 80 packets, 6599 bytes)
num  pkts bytes target    prot opt in     out     source               destination

(kali㉿kali)-[~/CIP-B103-Lab6]
$ sudo ip netns exec client curl -v \
--connect-timeout 5 \
http://192.168.44.128/firewall_lab.html
* Trying 192.168.44.128:80 ...

```



```
* Established connection to 192.168.44.128 (192.168.44.128 port 80) from 192.168.44.129 port 50580
* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 192.168.44.128
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Sat, 01 Aug 2026 16:08:17 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Sat, 01 Aug 2026 15:11:43 GMT
< ETag: "7e-657fdb90e4afd"
< Accept-Ranges: bytes
< Content-Length: 126
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: FUSEINI IMORU KANTUOGAA</p></body></html>
* Connection #0 to host 192.168.44.128:80 left intact

(kali㉿kali)-[~/CIP-B103-Lab6]
$
```

```
(kali㉿kali)-[~/CIP-B103-Lab6]
$ tree
.
├── evidence
│   ├── http_allowed.pcapng
│   └── http_blocked.pcapng
├── exported
├── reports
│   ├── allowed_vs_blocked.tsv
│   ├── apache_listener.txt
│   ├── http_allowed_sha256.txt
│   ├── iptables_after_add.txt
│   ├── iptables_after_test.txt
│   ├── iptables_before.rules
│   ├── iptables_before_sha256.txt
│   ├── iptables_before.txt
│   ├── iptables_restored.txt
│   ├── rule_verification.txt
│   ├── server_interfaces.txt
│   └── server_routes.txt
├── screenshots
├── scripts
└── working
```

7 directories, 14 files

```
(kali㉿kali)-[~/CIP-B103-Lab6]
```